

2.1 IRRATIONAL NUMBERS

2.1.1 Define an irrational number

We use numbers in our daily life. Almost every number of this world we can think of is a real number.

For example, 0, -1, 12.59, -0.3846, $\frac{3}{4}$, π are the real numbers.

Real numbers are a combination of rational and irrational numbers.

The numbers $\frac{2}{3}$, $\frac{1}{2}$, $\frac{5}{6}$ and $\frac{22}{7}$ represent the rational numbers. These numbers can be represented in the form of $\frac{p}{q}$ where p and $q \in \mathbb{Z}$ and $q \neq 0$. The set of rational numbers is denoted by $\mathbb{Q}$.

Irrational numbers cannot be expressed in the form of a ratio of integers as $\frac{p}{q}$, where $p, q \in \mathbb{Z}$ and $q \neq 0$. If we take square root of a number which is not a perfect square, it will be called an irrational number. $\sqrt{2}$, $\sqrt{3}$, $\sqrt{8}$, π are examples of irrational numbers. The Set of irrational numbers is denoted by $\mathbb{Q}'$.

Irrational numbers can be expressed in the form of decimal numbers which are non-terminating and non-recurring.

Irrational numbers are non-terminating and non-repeating decimals.

2.1.2 Recognize rational and irrational numbers

Let us recognize rational and irrational numbers with the help of examples: The numbers $\frac{2}{8}$, 4.755, 5, 0, $\sqrt{4}$, $2\sqrt{9}$, 1.9, 0.333..., $-\sqrt{49}$ and $3\frac{2}{3}$ are rational numbers.

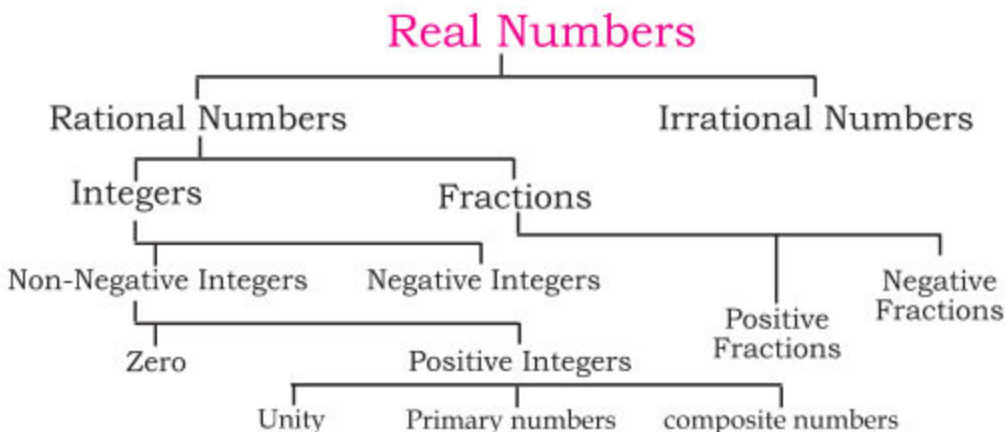
The numbers $\sqrt{\frac{2}{5}}$, $-3\sqrt{8}$, π , $\sqrt{7}$, $\sqrt{48}$, $\sqrt{122}$, are all irrational numbers.

$\sqrt{\frac{2}{5}}$ and $\sqrt{\frac{1}{3}}$ are irrational because in both, their numerator or denominator are not all integers.

2.1.3 Define real numbers

A real number is a value that shows any quantity mapped on a number line. Union of the sets of rational numbers ($\mathbb{Q}$) and irrational numbers ($\mathbb{Q}'$) is called the set of real numbers, denoted by $\mathbb{R}$, i.e. $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}'$.

We can conclude it by this diagram.



2.1.4 Demonstrate non-terminating/non-repeating (or non - periodic) decimals

Let us consider the following examples:

Example 1. Convert $\frac{1}{3}$ into decimal.

Solution:

$$\begin{array}{r}
 0.333\dots \\
 3 \overline{) 10} \\
 \underline{- 9} \\
 10 \\
 \underline{- 9} \\
 10 \\
 \underline{- 9} \\
 1
 \end{array}$$

So, $\frac{1}{3} = 0.333\dots$ or $0.\overline{3}$ is a non-terminating and repeating but periodic decimal (as 3 is repeating infinitely).

Note: If remainder $\neq 0$, the number is called non-terminating decimal.

Example 2. Convert $-\frac{4}{7}$ into decimal.

Solution:

$$\begin{array}{r}
 -0.571428 \\
 7 \overline{) -40} \\
 \underline{+35} \\
 -50 \\
 \underline{+49} \\
 -10 \\
 \underline{+7} \\
 -30 \\
 \underline{+28} \\
 -20 \\
 \underline{+14} \\
 -60 \\
 \underline{+54} \\
 -4
 \end{array}$$

So, $-\frac{4}{7} = -0.571428\dots$

Example 3. Convert $\frac{1}{2}$ and $-\frac{4}{5}$ into decimal.

Solution:

(i) $\frac{1}{2} = 0.5$, it is a terminating decimal.

(ii) $-\frac{4}{5} = -0.8$, it is also a terminating decimal.

Note: If remainder = 0 after division then common fractions is called terminating decimal fraction.

Hence every common fraction is either terminating or non-terminating but a repeating decimal. So every common fraction is a rational number.

Irrationals are numbers, whose digits in the decimal part are non-terminating and non-repeating in the same order, like $\sqrt{2} = 1.1412135\dots$, $\sqrt{5} = 2.23606\dots$, $\sqrt{243} = 15.5884\dots$ and $\pi = 3.1417\dots$, are called irrational numbers.

EXERCISE 2.1

A. Separate rational and irrational numbers from the following:

1. 35 2. $\sqrt{36}$ 3. $-2\frac{1}{40}$ 4. $2\sqrt{4}$ 5. 6π 6. $\frac{22}{7}$
 7. $\sqrt{\frac{2}{5}}$ 8. $-\sqrt{1}$ 9. $\frac{\pi}{5}$ 10. $\frac{\sqrt{2}}{\pi}$ 11. $\frac{\sqrt{9}}{\sqrt{25}}$ 12. $\sqrt{11}$

B. Identify terminating and non-terminating (periodic) decimal fractions from the following rational numbers.

1. $\frac{1}{7}$ 2. $\frac{2}{3}$ 3. 0 4. $\frac{1}{11}$ 5. $\frac{4}{3}$ 6. $-\frac{1}{5}$
 7. $-\frac{5}{6}$ 8. 2256 9. $\frac{9}{10}$ 10. $\frac{25}{26}$ 11. $\frac{243}{100}$ 12. $\frac{18750}{1000}$

2.2 SQUARES

The product of a number multiplied by itself is called its square. For example square of 2 is $2^2 = 2 \times 2 = 4$. In the same way square of a is $a^2 = a \times a$.

2.2.1 Find perfect square of a number

A natural number which is square of another natural number is called a perfect square.

For example $9 = 3^2$. So, 9 is a perfect square of 3.

Similarly, $16 = 4^2$. So, 16 is a perfect square of 4.

Example 1. Find the perfect squares of 11 and 65.

Solution: Perfect square of $11 = 11^2 = 11 \times 11 = 121$

Perfect square of $65 = 65^2 = 65 \times 65 = 4225$.

Note: We can find perfect squares of rational numbers but we cannot find perfect squares of irrational numbers.

For example. $(\sqrt{2})^2 = 2$ is a square but not a perfect square.

$(\pi)^2 = \pi^2$ is not a perfect square.

2.2.2 Establish patterns for the squares of natural numbers

Let us establish a pattern of adding natural numbers to get squares of natural numbers.

The first natural number is 1, therefore $1^2 = 1 \times 1 = 1$

Now $2^2 = 2 \times 2 = 4 = 2 + 1 + 1$ or $2^2 = 1 + 2 + 1$ (by adding current number to be squared with twice the number before it).

In the same way: $3^2 = 3 \times 3 = 9 = 3 + 2 + 2 + 1 + 1$ or $3^2 = 1 + 2 + 3 + 2 + 1$

Hence the desired pattern of squares by addition is as under:

$$\begin{aligned}
 1^2 &= && 1 &= 1 \\
 2^2 &= && 1 + 2 + 1 &= 4 \\
 3^2 &= && 1 + 2 + 3 + 2 + 1 &= 9 \\
 4^2 &= && 1 + 2 + 3 + 4 + 3 + 2 + 1 &= 16 \\
 5^2 &= && 1 + 2 + 3 + 4 + 5 + 4 + 3 + 2 + 1 &= 25 \\
 6^2 &= && 1 + 2 + 3 + 4 + 5 + 6 + 5 + 4 + 3 + 2 + 1 &= 36 \\
 7^2 &= && 1 + 2 + 3 + 4 + 5 + 6 + 7 + 6 + 5 + 4 + 3 + 2 + 1 &= 49 \\
 8^2 &= && 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 &= 64 \\
 9^2 &= && 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 &= 81 \\
 10^2 &= && 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 &= 100
 \end{aligned}$$

and so on...

We can also get squares of natural numbers by taking the sum of first “k” odd natural numbers, where “k” is that natural number whose square is needed.

$$1^2 = 1 = 1$$

$$2^2 = 1 + 3 = 4 \text{ (sum of first two odd numbers)}$$

$$3^2 = 1 + 3 + 5 = 9 \text{ (sum of first three odd numbers)}$$

$$4^2 = 1 + 3 + 5 + 7 = 16 \text{ (sum of first four odd numbers)}$$

$$5^2 = 1 + 3 + 5 + 7 + 9 = 25 \text{ (sum of first five odd numbers)}$$

$$6^2 = 1 + 3 + 5 + 7 + 9 + 11 = 36 \text{ (sum of first six odd numbers)}$$

$$7^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 = 49 \text{ (sum of first seven odd numbers) and so on...}$$

EXERCISE 2.2

A. Find perfect squares of the following:

- | | | | | |
|-------|-------|-------|-------|---------|
| 1. 5 | 2. 13 | 3. 19 | 4. 39 | 5. 45 |
| 6. 58 | 7. 63 | 8. 79 | 9. 97 | 10. 108 |

B. Find all the perfect squares between:

1. 10 and 30 2. 20 and 50 3. 40 and 70 4. 70 and 90

C. Write pattern of additions for the following squares:

- | | | | | |
|----------|-----------|-----------|-----------|------------|
| 1. 7^2 | 2. 9^2 | 3. 15^2 | 4. 11^2 | 5. 13^2 |
| 6. 8^2 | 7. 12^2 | 8. 16^2 | 9. 20^2 | 10. 27^2 |

2.3 SQUARE ROOTS

To find the square root is an operation opposite to find the perfect square of a number. For example, perfect square of 4 is 16 as $4^2 = 16$, therefore square root of 16 is 4.

We use the symbol $\sqrt{\quad}$ for square root. Thus, $\sqrt{16} = 4$.

Note: The square root of a positive number is a value which when multiplied by itself, gives the original number.

2.3.1 Find the square root of:

(i) a natural number (ii) a common fraction (iii) a decimal
given in perfect square form, by prime factorization and division method:

We have learnt methods of prime factorization and division to find the square root of a number in the previous class. Let us practice them.

(i) Square root of a natural number

Example 1. Find the square root of the following by prime factorization and division methods: **(i) 625** **(ii) 1600**

Solution: **(i)** Square root of 625

Prime factorization method

$$\begin{array}{r}
 5 \overline{) 625} \\
 \underline{5 } 125 \\
 5 25 \\
 \underline{5 } 5 \\
 1
 \end{array}$$

$$625 = 5 \times 5 \times 5 \times 5$$

$$\begin{aligned}
 \text{or } \sqrt{625} &= \sqrt{5 \times 5 \times 5 \times 5} \\
 &= 5 \times 5 = 25
 \end{aligned}$$

Division method

$$\begin{array}{r}
 25 \\
 2 \overline{) 625} \\
 \underline{4 } 225 \\
 45 \overline{) 225} \\
 \underline{20 } 25 \\
 5 \overline{) 25} \\
 \underline{20 } 5 \\
 5 \overline{) 5} \\
 \underline{5 } 0
 \end{array}
 \quad
 \begin{aligned}
 1 \times 1 &= 1 \\
 2 \times 2 &= 4 \\
 3 \times 3 &= 9 \\
 44 \times 4 &= 176 \\
 45 \times 5 &= 225 \\
 46 \times 6 &= 276
 \end{aligned}$$

$$\text{So, } \sqrt{625} = 25$$

Solution: **(ii)** Square root of 1600

Prime factorization method

$$\begin{array}{r}
 2 \overline{) 1600} \\
 \underline{2 } 800 \\
 2 \overline{) 800} \\
 \underline{2 } 400 \\
 2 \overline{) 400} \\
 \underline{2 } 200 \\
 2 \overline{) 200} \\
 \underline{2 } 100 \\
 2 \overline{) 100} \\
 \underline{2 } 50 \\
 5 \overline{) 50} \\
 \underline{5 } 25 \\
 5 \overline{) 25} \\
 \underline{5 } 5 \\
 5 \overline{) 5} \\
 \underline{5 } 0
 \end{array}$$

$$\text{So, } 1600 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 5$$

$$\begin{aligned}
 \text{or } \sqrt{1600} &= \sqrt{2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 5} \\
 &= 2 \times 2 \times 2 \times 5 = 40
 \end{aligned}$$

Division method

$$\begin{array}{r}
 40 \\
 4 \overline{) 1600} \\
 \underline{16 } 00 \\
 8 \overline{) 00}
 \end{array}
 \quad
 \begin{aligned}
 2 \times 2 &= 4 \\
 3 \times 3 &= 9 \\
 4 \times 4 &= 16
 \end{aligned}$$

Single '0' from the pair of zeros is shifted to square root

$$\text{So, } \sqrt{1600} = 40$$

(ii) The square root of a common fraction (when numerator and denominator are perfect squares)

Example 2. Find the square roots of the following by prime factorization and division methods: **(i)** $\frac{9}{16}$ **(ii)** $\frac{49}{64}$

Solution: **(i)** Square root of $\frac{9}{16}$

Prime factorization method

Numerator: 9

$$\begin{array}{r|l} 3 & 9 \\ 3 & 3 \\ \hline & 1 \end{array} \quad 9 = 3 \times 3$$

$$\text{or } \sqrt{9} = \sqrt{3 \times 3} = 3$$

Denominator: 16

$$\begin{array}{r|l} 2 & 16 \\ 2 & 8 \\ 2 & 4 \\ 2 & 2 \\ \hline & 1 \end{array} \quad 16 = 2 \times 2 \times 2 \times 2$$

$$\sqrt{16} = \sqrt{2 \times 2 \times 2 \times 2}$$

$$\sqrt{16} = 2 \times 2 = 4$$

$$\text{Therefore } \sqrt{\frac{9}{16}} = \frac{\sqrt{9}}{\sqrt{16}} = \frac{3}{4}$$

Division method

Numerator: 9

$$\begin{array}{r} 3 \\ 3 \overline{) 9} \\ \underline{+ 3} \\ 6 \end{array} \quad \begin{array}{l} 1 \times 1 = 1 \\ 2 \times 2 = 4 \\ \boxed{3 \times 3 = 9} \end{array}$$

$$\text{So, } \sqrt{9} = 3$$

Denominator: 16

$$\begin{array}{r} 4 \\ 4 \overline{) 16} \\ \underline{+ 4} \\ 8 \end{array} \quad \text{or } \sqrt{16} = 4$$

Solution: **(ii)** Square root of $\frac{49}{64}$

Prime factorization method

Numerator: 49

$$\begin{array}{r|l} 7 & 49 \\ 7 & 7 \\ \hline & 1 \end{array} \quad 49 = 7 \times 7$$

$$\sqrt{49} = \sqrt{7 \times 7}$$

$$\sqrt{49} = 7$$

Denominator: 64

$$\begin{array}{r|l} 2 & 64 \\ 2 & 32 \\ 2 & 16 \\ 2 & 8 \\ 2 & 4 \\ 2 & 2 \\ \hline & 1 \end{array} \quad 64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

$$\text{or } \sqrt{64} = \sqrt{2 \times 2 \times 2 \times 2 \times 2 \times 2}$$

$$= 2 \times 2 \times 2 = 8$$

Division method

Numerator: 49

$$\begin{array}{r} 7 \\ 7 \overline{) 49} \\ \underline{+ 7} \\ 14 \end{array} \quad \sqrt{49} = 7$$

Denominator: 64

$$\begin{array}{r} 8 \\ 8 \overline{) 64} \\ \underline{+ 8} \\ 16 \end{array} \quad \begin{array}{l} 6 \times 6 = 36 \\ 7 \times 7 = 49 \\ \boxed{8 \times 8 = 64} \end{array}$$

$$\text{or } \sqrt{64} = 8$$

$$\text{Therefore } \sqrt{\frac{49}{64}} = \frac{\sqrt{49}}{\sqrt{64}} = \frac{7}{8}$$

(iii) The square root of a decimal (Perfect Square Form)

Example 3. Find the square root of the following by prime factorization and division methods: (i) 0.01 (ii) 1.21 (iii) 0.64

Solution: (i) To find square root of 0.01

(a) Prime factorization method

$$0.01 = \frac{0.01}{100} = \frac{1}{100}$$

$$\text{Numerator} = 1, \sqrt{1} = 1$$

Denominator : 100

$$\begin{array}{r|l} 2 & 100 \\ 2 & 50 \\ 5 & 25 \\ 5 & 5 \\ & 1 \end{array} \quad \begin{array}{l} 100 = 2 \times 2 \times 5 \times 5 \\ \sqrt{100} = \sqrt{2 \times 2 \times 5 \times 5} \\ = 2 \times 5 = 10 \end{array}$$

$$\text{Therefore } \sqrt{0.01} = \sqrt{\frac{1}{100}} = \frac{\sqrt{1}}{\sqrt{100}} = \frac{1}{10} = 0.1$$

(b) Division method

$$\begin{array}{r} 0.1 \\ 1 \overline{) \hat{0.01}} \\ + 1 \quad 1 \\ \hline 2 \quad 0 \end{array}$$

$$\text{So, } \sqrt{0.01} = 0.1$$

Solution: (ii) To find square root of 1.21

Prime factorization method

$$1.21 = \frac{1.21}{100} = \frac{121}{100}$$

Numerator : 121

$$\begin{array}{r|l} 11 & 121 \\ 11 & 11 \\ & 1 \end{array} \quad \begin{array}{l} 121 = 11 \times 11 \\ \text{or } \sqrt{121} = \sqrt{11 \times 11} = 11 \end{array}$$

Denominator : 100

$$\begin{array}{r|l} 2 & 100 \\ 2 & 50 \\ 5 & 25 \\ 5 & 5 \\ & 1 \end{array} \quad \begin{array}{l} 100 = 2 \times 2 \times 5 \times 5 \\ \sqrt{100} = \sqrt{2 \times 2 \times 5 \times 5} \\ = 2 \times 5 = 10 \end{array}$$

$$\text{Therefore } \sqrt{1.21} = \sqrt{\frac{121}{100}} = \frac{\sqrt{121}}{\sqrt{100}} = \frac{11}{10} = 1.1$$

Division method

$$\begin{array}{r} 1.1 \\ 1 \overline{) \hat{1.21}} \\ + 1 \quad 1 \\ \hline 21 \quad 021 \\ + 1 \quad 21 \\ \hline 22 \quad 0 \end{array}$$

$$\text{So, } \sqrt{1.21} = 1.1$$

(iii) To find square root of 0.64 (Students will solve it themselves)

EXERCISE 2.3

A. Find the square root of the following by prime factorization method.

1. 961 2. 1296 3. 344569 4. 817216
 5. $\frac{81}{121}$ 6. $\frac{2129}{9604}$ 7. $\frac{1764}{7744}$ 8. $1\frac{984}{14641}$
 9. 249.64 10. 0.4096 11. 2981.16 12. 131.1025

B. Find the square root of the following by division method.

1. 1029 2. 14161 3. 996004 4. 10329796
 5. $\frac{169}{289}$ 6. $\frac{1225}{2809}$ 7. $\frac{29241}{55696}$ 8. $1\frac{1089}{1936}$
 9. 648.7209 10. 180.9025 11. 727.9204 12. 7613.609536

2.3.2 Find the square root of a number which is not a perfect square

All natural numbers and rational numbers are not perfect squares.

For example, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{6.4}$, $\sqrt{2.5135}$.

Division method is used to find the square root of such numbers upto required decimal places. These square roots are irrational numbers.

Example 1. Find the square root of 3 up to two decimal places.

Solution: As there is no decimal value in 3, so we put two pairs of zeros to get the required decimal places.

Therefore, $\sqrt{3} = 1.73$

(Upto two decimal places)

1.73	3.0000	
1	1	
+ 1	200	26 × 6 = 156
27	189	(27 × 7 = 189)
+ 7	1100	28 × 8 = 224
343	1029	342 × 2 = 684
+ 3	71	(343 × 3 = 1029)
		344 × 4 = 1376

Example 2. Find the square root of 40.13 upto three decimal places.

Solution:

6.334	40.130000	
6	36	
+ 6	413	122 × 2 = 244
123	369	(123 × 3 = 369)
+ 3	4400	124 × 4 = 496
1263	3789	1262 × 2 = 2524
+ 3	61100	(1263 × 3 = 3789)
12664	50656	1264 × 4 = 5056
+ 4	10444	12663 × 3 = 37989
12668		(12664 × 4 = 50656)
		12665 × 5 = 63325

Therefore, $\sqrt{40.13} = 6.334$
 (upto 3 decimal places)

EXERCISE 2.4

A. Find the square root of the following upto 2 decimal places.

1. 2 2. 5 3. 13 4. 141 5. 180
 6. 2.5 7. 152.7696 8. 0.0143 9. 23.085 10. 125.08

B. Find the square root of the following upto 3 decimal places.

1. 6 2. 8 3. 11 4. 155 5. 205
 6. 7.3 7. 125.79 8. 1468.9 9. 129048.27 10. 3355.30

2.3.3 Use rule to determine the number of digits in the square root of a perfect square.

Number of digits in the square root of a perfect square can be calculated without finding the square root by the following rule:

Let n be number of digits in perfect square, then its square root contains: (i) $\frac{n}{2}$ digits when n is even and

(ii) $\frac{n+1}{2}$ digits when n is odd.

Example 1. Find the number of digits in the square root of 240100 and verify your answer.

Solution: As number of digits in 240100 is 6 which is an **even** number. So, number of digits in the square root of 240100 will be: $\frac{n}{2} = \frac{6}{2} = 3$

Hence, $\sqrt{240100} = 490$ which has three digits.

Verification:

$$\begin{array}{r}
 490 \\
 4 \overline{) 240100} \\
 \underline{+ 4 } 16 \\
 89 \overline{) 801} \\
 \underline{+ 9 } 801 \\
 98 \overline{) 00}
 \end{array}$$

Example 2. Find the number of digits in the square root of 34596 and verify your answer.

Solution: As number of digits in 34596 is 5 which is **odd**. So number of digits in the square root of 34596 will

$$\text{be: } \frac{n+1}{2} = \frac{5+1}{2} = \frac{6}{2} = 3$$

Hence, $\sqrt{34596} = 186$ which has 3 digits.

Verification:

	186
1	34596
+ 1	1
28	245
+ 8	224
366	2196
+ 6	2196
372	0

EXERCISE 2.5

Find the number of digits in the square roots of the following perfect squares.

- | | | | |
|-------------|--------------|----------------|---------------|
| 1. 8649 | 2. 250000 | 3. 322624 | 4. 614656 |
| 5. 432964 | 6. 81018001 | 7. 43388569 | 8. 3474748809 |
| 9. 73547776 | 10. 33942276 | 11. 7913169936 | 12. 935566751 |

2.3.4 Solve real life problems involving square roots

Example 1. In a room 1089 marbles are arranged on the floor in such a way that the number of marbles in each row is the same as the number of rows. Find how many rows of marbles are there.

Solution: Since the number of marbles in each row is the same as the number of rows, we have to find the square root of 1089.

$$\text{As } \sqrt{1089} = 33$$

So, there are 33 rows of marbles.

	33
3	1089
+ 3	9
63	189
+ 3	189
66	0

Example 2. The area of a square field is 4225 square metres. Find the length of its side.

Solution: As area of a square = (side)²

$$\text{So, Side} = \sqrt{\text{Area of square field}} = \sqrt{4225}$$

$$\text{As } \sqrt{4225} = 65$$

Therefore, the length of a side = 65 metres.

$$\begin{array}{r} 65 \\ 6 \overline{) 4225} \\ \underline{+ 6} 36 \\ 125 625 \\ \underline{+ 5} 625 \\ 130 0 \end{array}$$

Example 3. Find the least number of students:

(i) When subtracted from 124

(ii) When added to 124,

(iii) When multiplied by 124, in order to make it a perfect square

$$\begin{array}{r} 11 \\ 1 \overline{) 124} \\ \underline{+ 1} 1 \\ 21 24 \\ \underline{+ 1} 21 \\ 22 3 \end{array}$$

Solution: Let us find the square root of the non-perfect square number 124.

$$\text{As } 124 = 11^2 + 3 \quad (\text{greater than } 11^2 \text{ by } 3)$$

(i) Hence the required least number is 3 which when subtracted from 124, i.e., $124 - 3 = 121$, gives a perfect square (11^2 by 11)

(ii) As $124 = 12^2 - 20$, So 20 is the required least number which when added to 124 i.e. $124 + 20 = 144$, gives a perfect square (12^2)
Thus $124 + 20 = 12 \times 12 = 144$, which is the perfect square.

(iii)

$$\begin{array}{r} 2 \overline{) 124} \\ \underline{2} 62 \\ 31 31 \\ \underline{ 31} 0 \end{array}$$

Thus, $124 = 2 \times 2 \times 31$. As 31 is not occurring in pair, so 31 is the required least number by which when 124 is multiplied i.e. $2 \times 2 \times 31 \times 31$ gives a perfect square $(2 \times 31)^2$.

EXERCISE 2.6

1. A contribution was made for a party of students. Each student contributed as much as the number of students present. The collected amount was Rs 8,464. Find the amount paid by each.
2. The area of a square field is 10816m^2 . Find the length of its side.
3. In a school assembly 1129 students were standing in rows such that the number of rows are equal to the number of students in a row. Find the number of students to leave to make a complete square of rows.
4. The area of a rectangular field is 1,832 square metres. Its breadth is half of its length. Find the length of the rectangular field.
5. The area of a square field is 390625 square metres. Find the length of wire required for fixing along the sides as a fence.

2.4 CUBES AND CUBE ROOTS

2.4.1 Recognize cubes and perfect cubes

(i) Cube: The cube of a number is the number obtained by raising it to the power 3. It means the number is multiplied three times by itself.

Examples. $1 \times 1 \times 1 = 1^3 = 1$, $2 \times 2 \times 2 = 2^3 = 8$

$$3 \times 3 \times 3 = 3^3 = 27, \quad 4 \times 4 \times 4 = 4^3 = 64$$

$$5 \times 5 \times 5 = 5^3 = 125, \quad 6 \times 6 \times 6 = 6^3 = 216.$$

This means: 1 is the cube of 1, 8 is the cube of 2, 27 is the cube of 3, 64 is the cube of 4, 125 is the cube of 5 and 216 is the cube of 6.

(ii) Perfect Cube: It is a number that is the result of multiplying an integer by itself three times, or it is an integer to the third power of an integer.

Examples. $(7)^3 = 7 \times 7 \times 7 = 343$, $8^3 = 8 \times 8 \times 8 = 512$

$$(9)^3 = 9 \times 9 \times 9 = 729, \quad 10^3 = 10 \times 10 \times 10 = 1000$$

Hence 1, 8, 27, 64, 125, 216, 343, 512, 729 and 1000 are perfect cubes of 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10 respectively.

Similarly, we can find perfect cubes of rational numbers, like 1.1, 2.5 etc.

Perfect cube of 1.1 is $(1.1)^3 = (1.1) \times (1.1) \times (1.1) = 1.331$

Perfect cube of 2.5 is $(2.5)^3 = (.5) \times (2.5) \times (2.5) = 15.625$.

Note: Perfect cubes of irrational numbers do not exist.

Example 1. Examine whether these are perfect cubes or not.

(i) 1728 (ii) $5 \frac{23}{64}$ (iii) 13.824

Solution: We can examine by making their prime factors. First of all, we find the prime factorization as under.

$$\begin{aligned} \text{(i) } 1728 &= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \\ &= 1728 = 2^3 \times 2^3 \times 3^3 = (2 \times 2 \times 3)^3 = 12^3 \end{aligned}$$

Thus, 1728 is perfect cube of 12.

$$\text{(ii) } 5 \frac{23}{64} = \frac{5 \times 64 + 23}{64} = \frac{320 + 23}{64} = \frac{343}{64}$$

$$\frac{343}{64} = \frac{7 \times 7 \times 7}{4 \times 4 \times 4} = \left(\frac{7}{4}\right)^3 = \left(1 \frac{3}{4}\right)^3$$

Hence, $5 \frac{23}{64}$ is perfect cube of $\left(1 \frac{3}{4}\right)$.

$$\text{(iii) } 13.824 = \frac{13.824}{1000} = \frac{13824}{1000}$$

$$\begin{aligned} \frac{13824}{1000} &= \frac{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (3 \times 3 \times 3)}{(2 \times 2 \times 2) \times (5 \times 5 \times 5)} \\ &= \frac{2^3 \times 2^3 \times 2^3 \times 3^3}{2^3 \times 5^3} \\ &= \frac{(2 \times 2 \times 2 \times 3)}{(2^3 \times 5^3)} = \left(\frac{24}{10}\right)^3 = (2.4)^3 \end{aligned}$$

Hence, $13.824 = (2.4)^3$

or 13.824 is perfect cube of 2.4

2	1728
2	864
2	432
2	216
2	108
2	54
3	27
3	9
3	3
	1

Numerator

2	13824
2	6912
2	3456
2	1728
2	864
2	432
2	216
2	108
2	54
3	27
3	9
3	3
	1

Denominator

2	1000
2	500
2	250
5	125
5	25
5	5
	1

EXERCISE 2.7

A. Find the perfect cubes of the following:

1. 10 2. 20 3. 30 4. 40 5. 50 6. 60
7. 8 8. 12 9. 15 10. 18 11. 24 12. 25

B. Find the cubes of the following fractions:

1. $\frac{1}{2}$ 2. $\frac{2}{3}$ 3. $\frac{3}{4}$ 4. $\frac{4}{5}$ 5. $1\frac{1}{3}$ 6. $2\frac{1}{2}$
7. $1\frac{1}{4}$ 8. $2\frac{1}{6}$ 9. $1\frac{1}{10}$ 10. $2\frac{3}{20}$ 11. $1\frac{1}{9}$ 12. $3\frac{1}{3}$

C. Find the cubes of the decimals:

1. 0.4 2. 0.9 3. 1.2 4. 1.6 5. 2.1 6. 2.5
7. 0.01 8. 0.02 9. 0.05 10. 1.12 11. 1.05 12. 1.01

D. Examine whether the following are perfect cubes or not.

1. 512 2. 729 3. 3,375 4. 1,728 5. 64 6. 125 7. 4,096
8. 5,832 9. 9,261 10. 27,000 11. 32,768 12. 39,304 13. 1,728

2.4.2 Find the cube roots of numbers which are perfect cubes.

We know that, $8 = 2 \times 2 \times 2 = (2)^3$. It means cube of 2 is 8, i.e., $2^3 = 8$ or we can say cube-root of 8 is 2.

It can be written as $\sqrt[3]{8} = 2$ or $(8)^{\frac{1}{3}} = 2$.

Similarly, we can find the cube root of any perfect cube with the help of prime factorization.

Example. Find the cube root of 1. 1000 2. 216 3. 729

Solution: (1) $1000 = 10 \times 10 \times 10 = 10^3$

Therefore, $\sqrt[3]{1000} = 10$ or $(1000)^{\frac{1}{3}} = 10$.

Solution: (2) $216 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 = (2 \times 2 \times 2) \times (3 \times 3 \times 3)$

$$216 = 2^3 \times 3^3 = 6^3$$

Therefore $\sqrt[3]{216} = 6$ or $(216)^{\frac{1}{3}} = 6$

(3) $729 = (3 \times 3 \times 3) \times (3 \times 3 \times 3)$

$$= 3^3 \times 3^3 = 9^3$$

So, $\sqrt[3]{729} = 9$ or $(729)^{\frac{1}{3}} = 9$

2	216	3	729
2	108	3	243
2	54	3	81
3	27	3	27
3	9	3	9
3	3	3	3
	1		1

2.4.3 Recognize Properties of Cubes of Numbers

Following are some recognized properties of cubes of numbers with examples.

(i) Cube of a positive number is always positive.

For example $2^3 = 8$, $5^3 = 125$.

(ii) Cube of a negative number is always negative.

For example $(-3)^3 = -27$, $(-6)^3 = -216$.

(iii) Cube of an even number is always an even number e.g., $2^3 = 8$, $4^3 = 64$.

(iv) Cube of an odd number is always an odd number,

For example $3^3 = 27$, $5^3 = 125$.

(v) Cubes satisfy the distributive properties as under:

(a) multiplication and (b) division

(a) $(2 \times 3)^3 = 2^3 \times 3^3$, $(8 \times 9)^3 = 8^3 \times 9^3$ **(b)** $\left(\frac{4}{5}\right)^3 = \frac{4^3}{5^3}$, $\left(\frac{6}{7}\right)^3 = \frac{6^3}{7^3}$

(vi) Cube numbers are the perfect cubes. $7^3 = 343$, $9^3 = 729$

So, 343, 729 and 1728 are perfect cubes

(vii) Every perfect cube is a cube.

EXERCISE 2.8

A. Find the cube roots of the following:

1. 64 2. 343 3. 1331 4. 512 5. 729

6. 74088 7. 3375 8. 13824 9. 15625 10. 35937

11. $\frac{1}{8}$ 12. $2\frac{10}{27}$ 13. $1\frac{61}{64}$ 14. $1\frac{91}{125}$ 15. $\frac{343}{729}$

B. Verify with examples that:

- (i) The cube of a positive number is always a positive number.
- (ii) The cube of a negative number is always a negative number.
- (iii) The cube of every even number is always an even number.
- (iv) The cube of every odd number is always an odd number.

C. Verify the following:

- 1. $(3 \times 4)^3 = 3^3 \times 4^3$
- 2. $\left(\frac{2}{3}\right)^3 = \frac{2^3}{3^3}$
- 3. $(2 \times 5)^3 = 2^3 \times 5^3$
- 4. $\left(\frac{3}{4}\right)^3 = \frac{3^3}{4^3}$
- 5. $(4 \times 5)^3 = 4^3 \times 5^3 = 4^3 \times 5^3$
- 6. $(10 \times 20)^3 = 10^3 \times 20^3$
- 7. $\left(\frac{5}{7}\right)^3 = \frac{5^3}{7^3}$
- 8. $\left(\frac{3}{5}\right)^3 = \frac{3^3}{5^3}$
- 9. $(5 \times 8)^3 = 5^3 \times 8^3$

REVIEW EXERCISE 2

- 1. Three options are given against each statement. Encircle the correct one.
 - (i) Real number is:
 - (a) difference of rational numbers and irrational numbers
 - (b) intersection of set of rational numbers and set of irrational numbers
 - (c) union of set of rational numbers and a set of irrational numbers
 - (ii) Which of the following is not true about 100?
 - (a) whole number (b) rational number (c) irrational number
 - (iii) Which one of the following is a perfect square?
 - (a) 121 (b) 1.21 (c) 12.1
 - (iv) Square of 4.9 is:
 - (a) 4.90 (b) 24.01 (c) 240.1

(v) $\left(\frac{3}{5}\right)^2 = ?$

(a) $\frac{9}{5}$

(b) $\frac{3}{25}$

(c) $\frac{9}{25}$

(vi) $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = ?$

(a) 8^2

(b) 9^2

(c) 7^2

(vii) If the side length of a square is $0.6m$ then its area is:

(a) $0.36m^2$

(b) $3.6m^2$

(c) $36m^2$

(viii) $\sqrt{.04} = ?$

(a) .02

(b) 2.0

(c) 0.2

(ix) $\sqrt{1^2 \times 4^2} = ?$

(a) 4

(b) 14

(c) 41

(x) $\sqrt[3]{125} = ?$

(a) 3

(b) 4

(c) 5

(xi) $\sqrt{\frac{4}{9}} = ?$

(a) $\frac{2}{3}$

(b) $\frac{2}{9}$

(c) $\frac{4}{3}$

(xii) $\sqrt{\frac{a}{b}} = ?$

(a) $\frac{a}{b}$

(b) $\sqrt{\frac{b}{a}}$

(c) $\frac{\sqrt{a}}{\sqrt{b}}$

2. Find the number of digits in the square root of the following numbers. Also find the square root.

(i) 12,544

(ii) 418,609

(iii) 30,349,081

3. Find the square root of the following:

(i) $17 \frac{128}{289}$

(ii) $28 \frac{4}{9}$

(iii) $101 \frac{92}{169}$

(iv) 0.053361

(v) 0.204304

(vi) 65.61

4. The area of a square field is $161604m^2$. Find the length of one of its side.
5. Find the cube root of the following numbers.

(i) 3375 (ii) 1728 (iii) $\frac{343}{512}$ (iv) $1\frac{602}{729}$

SUMMARY

- The number which cannot be written in the form $\frac{p}{q}$, where $p, q \in \mathbb{Z}$ and $q \neq 0$ is called an irrational number.
- Set of real numbers is the union of set of rational and set of irrational numbers, i.e., $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}'$.
- A number whose decimal representation is terminating is called rational number.
- The decimal fraction in which the number of digits after the decimal point is finite, is called terminating decimal fraction.
- The decimal fraction, in which the number of digits after the decimal point is infinite, is called non-terminating.
- The product of a number by itself is known as square of that number.
- Square of a positive integer or rational number is multiplied by itself two times.
- The square root of a positive number is an other positive number whose square is the given number.
- If n is the number of digits in a perfect square, then its square root contains (i) $\frac{n}{2}$ digits if n is even (ii) $\frac{n+1}{2}$ digits if n is odd.
- Cube of a number means the number is multiplied by itself three times.

- For any rational number

(i) $\sqrt[3]{\frac{a}{b}} = \frac{\sqrt[3]{a}}{\sqrt[3]{b}}$ (ii) $\sqrt[3]{ab} = \sqrt[3]{a} \times \sqrt[3]{b}$